(Please write your Enrollment Number) Enrollment No._____

**MID-TERM EXAMINATION**
**(B.Tech. CSE/ECE) 1st semester**
**(January, 2023)**
**OFF LINE mode**

**Subject Code:** BAS-105 **Subject:** APPLIED CHEMISTRY

**Time :** 1 ½Hours **Maximum Marks :** 30

**Q1 (Attempt all Parts)** **(2.5*4)**

Support your answer with relevant chemical reactions or diagrams (if any).

(a) What is Calgon conditioning? Why Calgon conditioning is better than phosphate conditioning? Explain with the help of chemical reactions

(b) Name the salts that contribute to the reduced usage of lime or soda during water purification process. Explain with a suitable chemical reaction.

(c) Determine the number of Components, phases present in the following systems

$NH_4Cl(g) \rightleftharpoons NH_3(g)+HCl\ (g)$

$CaCO_3(s) \xrightleftharpoons{Heat} CaO\ (s)+CO_2(g)$

(d) What is the advantage of catalyst over high temperature to promote chemical reactions? Why a rough surface piece of platinum metal acts as a good catalyst in comparison to a smooth surface in the manufacture of $H_2SO_4$ by contact process.

**(5,5)**

**Q2 (Attempt any Two Parts )**

(a) A water sample contains $CaSO_4$ = 45 mg/L, $MgCO_3$ = 50 mg/L, $CaCO_3$ =250 mg/L, $MgSO_4$=85 mg/L, $MgCl_2$=80 mg/L, $SiO_2$ = 290 mg/L. Calculate the amount of Lime and Soda required for softening25,000 liters of hard water per day for a year.

(b) Describe the 'Demineralization process' of softening of hard water. In Deionization process, water is usually first passed through the cation exchanger and then through the anion exchanger. Give reason.

(c) 200 mL of sample required 20 mL of N/50 HCl using methyl orange as indicator. Another 200 mL of same sample required 8 mL of N/50 HCl using phenolphthalein as indicator. Express the alkalinities in terms of mg $CaCO_3$ per liter.

**(5,5)**

**Q3 (Attempt any Two Parts )**

(a) What are acid base catalysts? Discuss the mechanism of acid catalyzed reaction with suitable example.

(b) Discuss the water system with the help of phase diagram. Label the 'Super cooled curve' in the diagram and explain the metastable state.

(c) Explain 'Adsorption theory' with a suitable example. Explain the role of molybdenum (Mo) in Haber's process? And the effect of 'oxides of arsenic' in Contact process? With suitable reactions of each they involve?